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Portfolio: https://portfolio-lu9d.onrender.com/ | Github: https://github.com/sukikatte

EDUCATION

Nanyang Technological University, Singapore Aug 2026 - Present

MSc Game Design, Art and Technology

Relevant Coursework: Principles of Game Design; Modelling & Animation for Games; Game AI; Art-Tech Collaboration

Upcoming Coursework: Immersive Environment Design; AI Driven Computer Graphics and Visualization; Games for Society

University College Dublin, Ireland & Beijing University of Technology, China Sept 2022 - Jun 2026

Dual Degree in Software Engineering - BSc (UCD) & BEng (BJUT) | Second Class Honours, Grade 1 | GPA: 3.49/4.2

Selected for UCD Exchange Programme for final year; awarded First-Class Overseas Study Scholarship

GAME DESIGN & DEVELOPMENT PROJECTS

Unity 3D Corridor Game: Keep Burning Jan 2026 - May 2026

Independent Development | Unity, Blender, C#

- Designed a corridor runner around a living flame whose vigor continuously drains, creating a resource-management gameplay loop in which players collect fuel, avoid water hazards, and burn through obstacles to reach the end camp.
- Implemented core gameplay systems including player movement, jump cost, vigor scoring, hazard/pickup logic, and dynamic corridor segment spawning; mapped vigor to flame scale, color, and lighting for readable gameplay feedback.

Java Desktop Rhythm Game: The Key Jan 2026 - May 2026

Independent Development | Java, Swing, Java MIDI

- Designed a dual-staff rhythm game that maps pitch to vertical position on treble and bass staves, featuring Free Compose and Song Performance modes with two-voice playback and live MIDI note output.
- Implemented MVC gameplay systems including dual-player note movement, mute/rest toggling, scrolling hit markers, and timing-and-pitch scoring; mapped note position to MIDI note values and rendered music-notation feedback for readable gameplay and accurate note input.

Honor of Kings Lineup Selection Agent Jan 2026 - May 2026

Independent Development | Python, OpenAI Responses API, RAG, Function Calling, REST APIs

- Designed a calm, reasoned HoK draft coach that takes ally/enemy picks and the user's lane, generating persuasive pick recommendations, counter/synergy analysis, win conditions, and teamfight roles.
- Built an LLM agent loop using RAG over curated hero/tactics data and function calling to live ranking APIs; added a numeric scoring layer for lane roles, synergy/counter weights, and small-sample adjustment before report generation, with Matplotlib visualizations of draft strength.

ADDITIONAL PROJECTS

Android Music Game: Guess Songs Sept 2024 - Jan 2025

Independent Development | Java, Android Studio, Firebase, Ali Cloud OSS

- Built a native Android app with modular MVVM architecture, featuring Classic/Timed/Multiplayer modes and a Material Design UI, and designed a responsive Material Design UI focused on usability and smooth navigation.
- Integrated Firebase Authentication, Realtime Database, and cloud storage to support real-time state sync, multiplayer room creation, user profiles, score history, and scalable music management and file storage.

Java Desktop Competitive Game: Mahjong Feb 2024 - Jun 2024

Team Development | Java 17, JavaFX 21, Maven, JUnit/TestNG

- Built a cross-platform Mahjong desktop game with MVC architecture, full rule engine, AI opponents, and real-time sync.
- Implemented advanced win detection, dynamic scoring, and a responsive JavaFX UI with event-driven interaction.

Dwen Dwen's Neighbour: China Animal Protection Platform Feb 2025 - Jun 2025

Team Development | Python, JavaScript, Flask, Firebase, Web Crawler

- Led the development of a bilingual platform supporting multi-user interaction, featuring an animal encyclopedia with web-scraped content, interactive games, virtual panda care (SVG), community forums, and donation features.
- Designed and implemented the backend with Flask and CORS policies, using Firebase Realtime Database for concurrent data handling and ECharts for interactive visualizations; deployed the system with Gunicorn and multi-worker architecture. Awarded 3rd Prize at the BJUT "Internet+" Innovation & Entrepreneurship Competition.

Hua Xia: Chinese Medicine Knowledge Platform Feb 2025 - Jun 2025

Team Development | Vue.js, JavaScript, Flask, SQLAlchemy, Fuzzy Matching

- Developed a bilingual full-stack platform with role-based access control, featuring a herb encyclopedia, prescription management, AI consultation chat, interactive storytelling, and distribution visualization.
- Utilized Flask/SQLAlchemy for backend services, Vue.js for the responsive frontend, JWT for authentication, and Gunicorn for production deployment with multi-worker architecture; 3rd Prize, BJUT "Internet+" Competition.

RESEARCH EXPERIENCE

Frankenstories: Story Generation from Reusable Parts Sept 2025 - May 2026

Final Year Project, UCD | Supervisor: Assoc. Prof. Tony Veale | Grade: A

- Developed an LLM-based comic narrative generation system that recombines pre-existing narrative elements within the ComiXML framework to create novel stories while maintaining content grounding.
- Designed a two-stage evaluation framework to assess structural validity and narrative quality, combining XML/schema validation and hallucination-rate analysis with LLM-as-a-judge assessment of coherence, continuity, and entertainment.
- Produced a research dissertation featuring rendered comic generations and practical guidelines for constrained generation.

Lucy Cavendish College, University of Cambridge, UK Jul 2023 - Aug 2023

Summer Research Programme in Artificial Intelligence and Computer Science | Score: 88/100

Developed a deep learning demo with a team to detect attentiveness in online classrooms through facial pose recognition.

INTERNSHIPS

Software Engineer Intern | SinoRail Information Engineering Group Jun 2025 - Aug 2025

- Prepared technical documentation and analyzed internal project data to support reviews, tracking, and decisions.

Project Management Intern | Earthtown Interior Design Studio Jul 2024 - Aug 2024

- Coordinated project schedules and deliverables, tracked action items, and prepared reports to support project execution.

EXTRACURRICULAR ACTIVITIES

Keyboardist and Lead Vocalist | Campus Band Sept 2022 - Feb 2025

- Served as a core member of the campus band, responsible for keyboard playing, vocals, and arrangements.
- Designed and produced 10+ posters and visual images for the band.

Scriptwriter & Director | Campus Musical Society Performance Sept 2022 - July 2024

- Scripted and directed *The Greatest Showman* musical, leading cast members and coordinating full-stage production.

SKILLS

Game Development: Unity, C#, Java, Python, JavaScript

AI & Backend: LLMs, RAG, Function Calling, Flask, Spring Boot, REST APIs, Firebase, MySQL, SQLite

Design & 3D Modeling: Figma, Maya, Blender, ZBrush

Development Tools: Git, Docker, Maven, Gradle, IntelliJ IDEA, PyCharm, Eclipse, Android Studio

LANGUAGES & INTERESTS

Mandarin (native), English (IELTS 7.5), French (Intermediate), Japanese (Basic)

Piano, Violin, Drawing, Singing, Skating